

Marking Scheme

1. Amylase is an enzyme that breaks down starch. A student investigated the effect of pH on amylase activity by using a starch agar plate. Six circular wells were cut into the agar plate.

Each well contained the same concentration and volume of amylase, and a buffer solution of different pH. The agar plate was then left for 24 hours. Figure 1.1 shows the results.

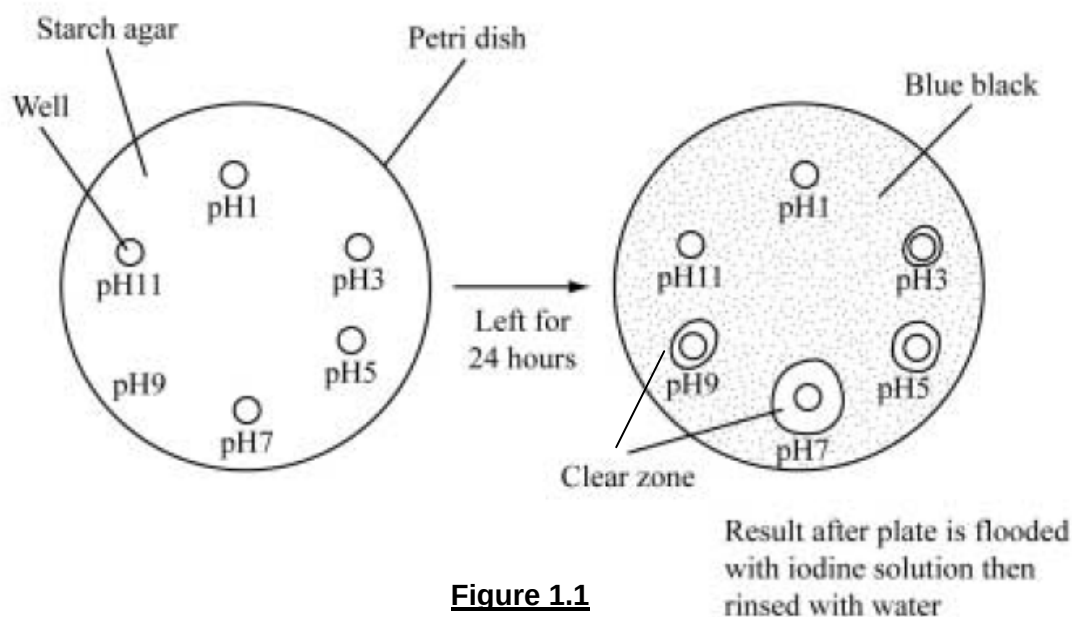


Figure 1.1

- (a) Describe how the student could have used these results to compare the activity of the enzyme at different pH values. [1]
 - **Measure diameter or radius (using ruler)/ area of clear zone (using graph paper);**
 - **Larger the area cleared/ bigger the diameter of clear zone would suggest higher activity of enzyme;**
- (b) The student concluded that the optimum pH for amylase activity was 7. This conclusion may not be valid. Explain why. [1]
 - **No measurements at intermediate pH values i.e. 5-7 / 7-9;;**
- (c) Explain how the substrate may be attached to the enzyme. [2]

Spatial fit:

- **Only allowing substrate with shape complementary to the shape of the active site of the enzyme to bind to the enzyme's active site;;**
- **Substrate and enzyme are held at correct angle and orientation to ensure successful collision to form enzyme-substrate complex;;**

OR

Chemical fit:

- **The enzyme interacts with the substrate via hydrophobic interactions, hydrogen bonds and ionic bonds;; (2 bonds = ;)**
- **The types of bonds formed depends on the nature of the R group of the amino acids that made up enzyme's active site;;**
- **These bonds help bind the substrate to the enzyme, as a temporary structure known as the enzyme-substrate complex is formed at the active site;;**

Marking Scheme

(d) Using your knowledge of enzyme structure, explain the result obtained at pH 11. [3]

- **At extreme pH, enzymes are denatured / tertiary structure altered;**
- **Ionic / hydrogen bonds holding the precise 3D conformation of the enzyme are broken;**
- **Active site conformation/shape is lost;**
- **Substrate cannot bind to active site;**
- **No enzyme-substrate complex;**
- **formed and therefore no products formed;**
- **This is observed in the absence of clear zone at pH11, indicating no products formed;**

(e) Describe a control that would be necessary for this investigation. [1]

- **Use of denatured / boiled enzyme/ distilled water;;**
- **(At all pH values)**

[Total: 8 marks]

Marking Scheme

2. The blue mussel, *Mytilus edulis*, is a bivalve mollusc that is found fixed to hard surfaces in both the open sea (high salinity) and in estuaries (low salinity) of the North Atlantic.

It maintains a water potential in its body similar to that of the surrounding medium by increasing or decreasing the intracellular levels of free amino acids. As amino acids are soluble, increasing intracellular levels of amino acids will lower the water potential of tissues in mussels.

The Lap gene codes for an enzyme which catalyses the release of amino acids from polypeptides in the tissues. The Lap⁹⁴ allele codes for a more active form of this enzyme.

Populations of mussels in the open sea have high frequencies of the Lap⁹⁴ allele at the Lap locus. In estuaries (low salinity), amino acids are removed from the body resulting in a loss of organic nitrogen by the mussels and increased mortality. There is a progressive reduction in the frequency of this allele resulting in the low frequencies characteristic of adult populations.

- (a) Account for the progressive reduction of mussels with the Lap⁹⁴ allele in the estuaries. [4]

- With Lap⁹⁴, there would be more amino acids in the tissues, therefore water potential would decrease ;;
- This will be followed by intake of water by osmosis ;;
- This will cause the cells to swell and burst;;
- Later loss of amino acids from the body results in lack of protein synthesis and growth ;;
- Failure to survive ;;
- And to reproduce to produce fertile and viable offspring / unable to pass down trait to the subsequent generation;;

[max 4]

- (b) Using the symbols **E** for the Lap⁹⁴ allele and **e** for normal allele, draw genetic diagrams to explain how two mussels with high levels of free amino acids could produce offspring with normal levels of amino acids. [6]

Parental phenotype	Mussels with high levels of free amino acids	Mussels with high levels of free amino acids
Parental genotype	Ee ;	Ee ;
Parental gametes	E e	E e
F ₁ genotype	EE Ee Ee ;	ee ;
F ₁ phenotype	Mussels with high levels of free amino acids;	Mussels with normal levels of amino acids;
F ₁ phenotypic ratio	3 (Mussels with high levels of free amino acids) :	1 (Mussels with normal levels of amino acids) ;

* Wrong parents to start with – 0.

* Phenotypic ratio 1 (high levels of aa) : 2 (high levels of aa) : 1 (low levels of aa) not accepted.

* Phenotypic ratio 3 (Lap⁹⁴ allele) : 1 (Normal allele) not accepted.

Marking Scheme

A student counted the number of mussels with high levels of free amino acids and those with normal levels of amino acids. Table 2 summarises his results.

Type of mussel	Total number
mussels with high levels of free amino acids	185
mussels with normal levels of amino acids	57

Table 2

(c) (i) Explain why the ratio is not exactly the same as those observed in (b). [2]

- **Random fertilisation;;**
- **Incorrect counting;;**
- **Adverse environmental conditions;;**
- **Large sample size required;;**

(ii) Suggest a statistical that the student should carry out to verify whether if the difference of his results and the expected result is significant? [1]

- **Carry out chi-square test;;**

[Total: 13 marks]

Marking Scheme

3. Figure 3.1 shows part of a DNA molecule.

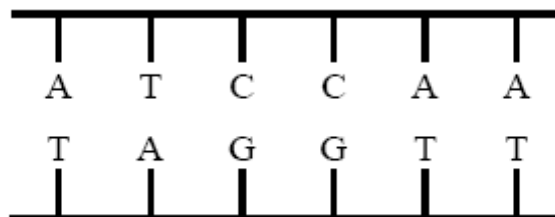
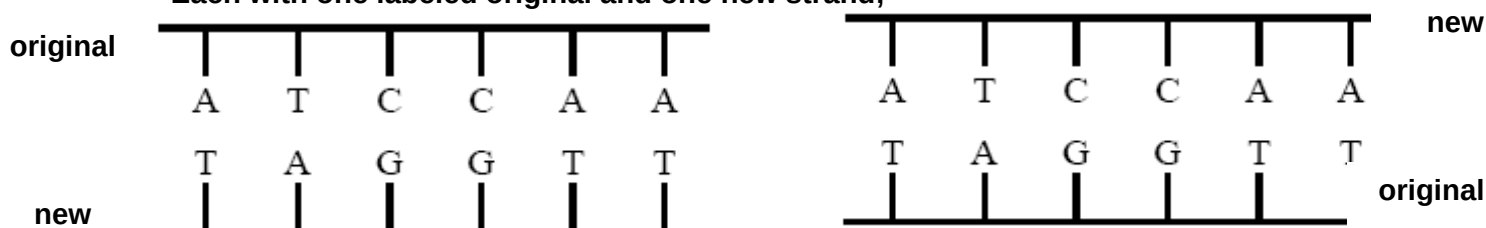


Figure 3.1

(a) In the space below, draw a similar diagram to show this part of the molecule after it has replicated. Label the original strands and the new strands. [2]

- **Diagram showing two identical molecules;**
- **Each with one labeled original and one new strand;**



(b) (i) State when does DNA replication occurs during mitosis. [1]

- **(S phase) of interphase;;**

(ii) DNA is replicated semi-conservatively. State what is meant by semi-conservative replication. [2]

- **the DNA strands separate first; by breaking hydrogen bonds;**
- **both strands act as templates;**
- **each new molecule of DNA containing one new and one original strand of DNA;**

(iii) Describe how the two original strands determine the structure of the two newly synthesized strands. [2]

- **(Free deoxyribonucleotides) will undergo complementary base pairing (with the bases) on the DNA template/ original strand;;**
- **A will base pair with T; G to C;**

Cytarabine is a drug used to treat cancer. The shape of a cytarabine molecule is very similar to that of a cytosine nucleotide but there are some differences. Cytarabine is incorporated into DNA.

(c) Suggest 2 reasons why DNA which contains cytarabine instead of cytosine cannot produce mRNA. [2]

- **RNA polymerase cannot bind to cytarabine;;**
- **Free ribonucleotides cannot recognise and bind to cytarabine;;**
- **No hydrogen bonds formed between complementary ribonucleotides and cytarabine;;**
- **Causes the termination of mRNA elongation;;**

[Total: 9 marks]

Marking Scheme

4. The bacterial plasmid, pBR322, was used as a vector to transform Gene X as shown in Figure 4.1 below.

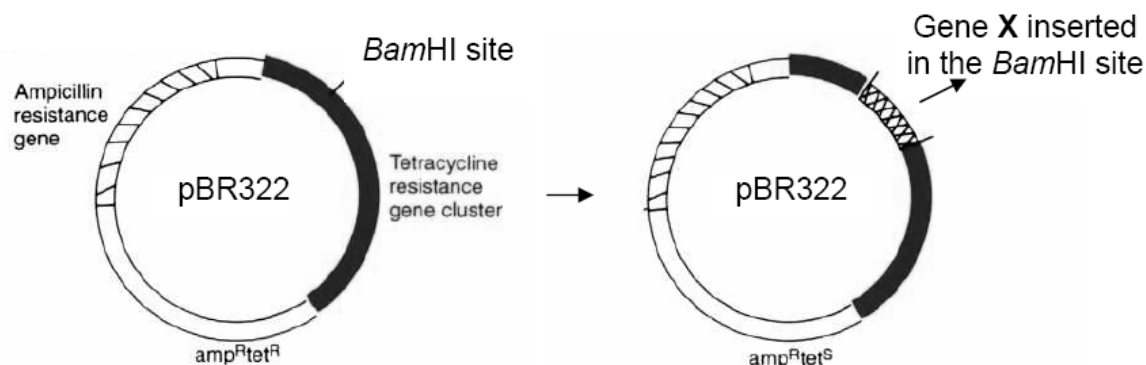


Figure 4.1

- (a) With reference to Figure 4.1, what is the significance of *Bam*HI? [2]
- **BamHI is a restriction enzyme that will recognise and cleave at specific recognition sequence/ site;**
 - **Via hydrolysis of phosphodiester bond;**
 - **Allow gene of interest to be inserted;**
 - **via complementary base pairing by hydrogen bonding;**
- (b) Restriction enzymes generate either sticky or blunt ends. What is the disadvantage of using restriction enzymes that generate blunt ends? [1]
- **Decrease efficiency of producing recombinant DNA molecules as;**
 - **Linker DNA required for reannealing;**

A tube of the prepared recombinant DNA molecules was then transformed into *Escherichia coli*. The transformed bacteria were plated onto nutrient agar plate containing ampicillin. Replica plating was subsequently carried out onto nutrient agar plate containing tetracycline. Bacterial growth on the plates is shown in Figure 4.2.

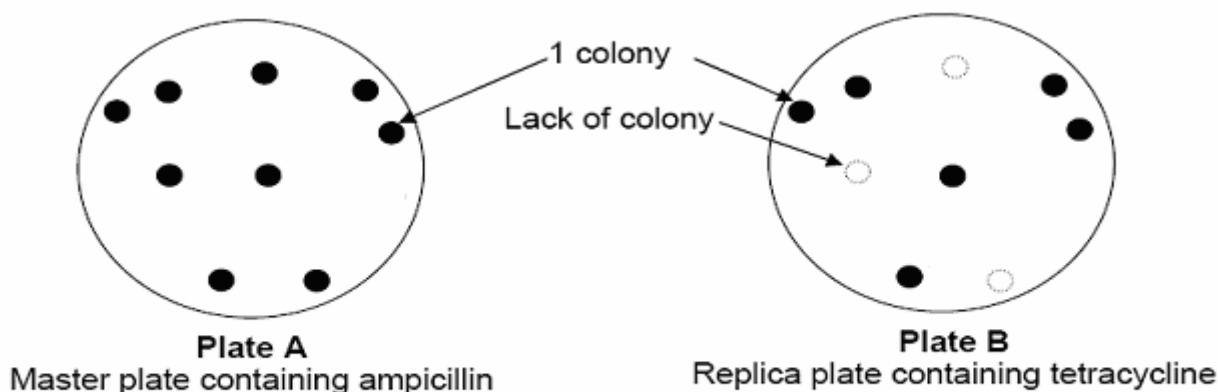


Figure 4.2

- (c) With reference to Figure 4.2,
- (i) State the plate that contains the transformed cells carrying the gene of interest [1]
- **A;;**

Marking Scheme

- (ii) Account for the difference in colony numbers. [3]
- Plate A selects for successfully transformed cells/ taken up plasmid with ampicillin gene;;/ recombinant plasmid
 - Plate B selects for non-recombinant cells/ taken up plasmid with tetracycline gene still intact;;/ non-recombinant plasmid
 - Therefore comparing the difference in colony numbers between Plate A and B will allow us to identify recombinant cells/ colonies;;
- (d) Describe the principle of insertional inactivation of the gene coding for resistance to tetracycline. [3]
- Tetracycline resistance gene codes for protein conferring resistance to tetracycline (antibiotics);
 - Non-recombinant cells will synthesize protein conferring resistance to antibiotics;
 - Colonies will grow on tetracycline plate;
 - Recombinant cells with tetracycline gene disrupted;
 - Proteins will not be produced;
 - Bacteria are killed/damaged by tetracycline;
 - Therefore colonies do not grow on tetracycline plate;

[3 max]

[Total: 10 marks]

Marking Scheme

5 (a) Describe how carbohydrates are formed in the Calvin cycle. [8]

- CO_2 combines with 5C sugar ribulose biphosphate (RuBP);;
- Catalysed by RuBP carboxylase/Rubisco;;
- 6C intermediate splits to form two 3C compound;;
- glycerate-3-phosphate (GP);;
- use of ATP;;
- reduction / use of NADPH;;
- to form 1 glyceraldehyde-3-phosphate (GALP/triose phosphate);;
- GALP/G3P used to regenerate RuBP;;
- utilising ATP molecules;;
- pairs of GALP/G3P/triose phosphate is used to synthesize hexose;;
- which is in turn used for synthesis of carbohydrate;;

(b) Outline the role of NAD^+ and NADP^+ in cells. [12]

NAD^+ and NADP^+

- Electron/hydrogen carrier or coenzymes;
- Carry hydrogen atoms from one compound to another;

NADP^+ in photosynthesis;

- Proton and electron carrier;
 - in non-cyclic photophosphorylation;
 - Drives photolysis of water;
 - $2\text{H}^+ + 2\text{e}^- + \text{NADP}^+ \rightarrow \text{NADPH}$ or reduced NADP
 - hydrogen ions; from splitting of water;
 - combine with NADP to form NADPH;
 - NADPH passes to Calvin cycle/ light independent reactions; in the stroma
 - NADPH reduces glycerate biphosphate/1,3 diphosphoglycerate; to
 - triose phosphate/ glyceraldehyde phosphate;
- and gets reoxidized back to NADP^+

NAD^+ in respiration;

Aerobic respiration:

- Glycolysis: Accept H^+ from triose phosphate/ glyceraldehyde phosphate;
 - which is oxidized to 1,3 diphosphoglycerate or pyruvate;
 - Forms 2 NADH for each glucose molecule;
 - Link reaction: Accept H^+ from pyruvate;
 - which is converted to acetyl CoA;
 - Forms 2 NADH for each glucose molecule;
 - Krebs cycle: Accept H^+ from intermediates in the regeneration of oxaloacetate from citrate;
 - Forms 6 NADH for each glucose molecule;
 - Link reaction and Krebs cycle occurs in the mitochondrial matrix;
 - Carry hydrogen/ electrons to ETC;
 - Oxidative phosphorylation/ transfer of electrons down the ETC to release ATP;
 - 1 NADH yields 3 ATP;
- NAD^+ reoxidised;

Anaerobic respiration:

- NAD^+ regenerated when electrons from NADH transferred to pyruvate/ ethanal;
- NAD^+ reused in glycolysis to generate 2 ATP;

Marking Scheme

6

(a) With reference to the analysis of DNA molecules, describe and explain the use of:

(i) Gel electrophoresis [4]

- DNA fragments placed/loaded in wells;
- At one end of an agarose gel;
- DNA negatively charged;
- Due to the presence of phosphate group;
- Moves towards the anode/positive electrode;
- Rate of movement/speed/distance travelled in a given time depends on size of DNA fragment;
- Use of DNA ladder/known lengths of DNA to calibrate;
- Use of loading dye;
- View under UV light to see DNA bands/AW;
- Allows identification of different alleles/mutations of genes/ other valid application;
- Allows sequencing of DNA;

(ii) The polymerase chain reaction (PCR) [8]

Denaturation

- 94-96°C (denaturation)
- Double stranded DNA separate/hydrogen bonds holding strands together break;
- Forming single stranded DNA;

Annealing

- 54-68 °C
- Complementary primer attaches to single stranded DNA/DNA template;
- Ref forward and reverse primers;
- Polymerase attaches (to the primer-template double stranded DNA) and starts copying the template/polymerising DNA/adding nucleotides;

Extension

- 72°C
- Complementary nucleotides join to forming strand/ extend strand attached to primer;
- Nucleotides attach to 3' end on both primers;
- by Taq polymerase
- Ref to automated thermal cycler;
- Amplify DNA;
- Specific gene (fragment);
- Much faster than previous methods using bacteria; Ref to tests for detection of bacteria/virus;
- Ref to determination of viral load/concentration;
- Ref to viral genotype determination/identification of specific viral clones;
- Ref to other named use of PCR (e.g. in, DNA fingerprinting/studying human ancestry/studying ancient DNA/paternity testing/cloning genes for using in genetic manipulation/making deliberate mutations in studying protein functioning/investigating gene expression)

Marking Scheme

- (b) Discuss the goals and implications of the human genome project, including the benefits and ethical concerns. [8]

Goals (max 3)

1. **To identify all the approximately 20,000 – 25,000 genes in human DNA /To obtain a genetic linkage map of human genome;**
2. **To determine the sequences of the 3 billion nucleotide base pairs of the entire human genome;**
3. **To store this information in databases;The Human Genome Project has developed user-friendly tools to provide scientists easy access to up-to-date information obtained from the project;**
4. **To improve tools for data analysis; Computer programmes have been developed to analyse the data to interpret genome information;**
5. **To transfer related technologies to the private sector; By licensing technologies to private companies, the project fostered the development of new medical applications;**
6. **To address the ethical, legal, and social issues (ELSI) that may arise from the project;**

Benefits

1. **Allows early detection of mutant genes, which causes disease, could make the diagnosis of certain diseases easier and could make it possible accurately to predict the onset of diseases;;**
2. **Provide insight into the types of protein encoded by these genes. Thus increase the potential to treat genetic and acquired diseases by gene therapy, whereby defective genes are either replaced by normal genes or supplemented with other genes to prevent a disease taking its course;;**
3. **Customised drugs can be designed based on genetic profile. Pharmaceutical products such as medicine can be synthesized using readily available gene sequences;;**
4. **Able to assess health damage and risks posed to individuals by exposure to radiation or carcinogens, in order to understand the effects of such agents on cancer risk;;**

Ethical issues (max 3)

1. **Ownership and use of personal genetic information.** Difficult to determine who owns and control the genetic information. Who should have access to it, are a matter for concern e.g. should employers/insurance agents have access to the genetic fingerprint and hire workers/granting insurance policies based on their genetic predisposition;;
2. **Reproductive issues regarding use of genetic information in reproductive decision making and reproductive rights.** There are concerns that prenatal genetic testing could lead to genetic manipulation or a decision to abort based on undesirable traits disclosed by the tests;;
3. **Clinical issues.** There are questions on the accuracy, reliability and utility of the genetic tests, how prepare are the healthcare professionals for the new genetics and how much can the interpretation be trusted;;

Marking Scheme

4. **Treatment versus Enhancement of humans.** There is no clear distinction between medical treatment and enhancement (i.e. to change the human genome to make it flawless);;
5. **Physiological impacts and stigmatisation due to an individual's genetic differences.** The information may affect the society's perceptions of an individual;;
6. **Commercialisation of products.** It is difficult to determine who own genes and other pieces of DNA and to ascertain if patenting DNA will limit the accessibility and development into useful products;;